

RAID (REDUNDANT ARRAY OF INDEPENDENT DISKS)

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AGENDA

Introduction

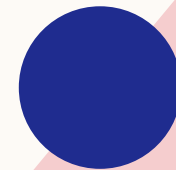
What is RAID?

RAID controller

RAID levels

Benefits of RAID

Summary

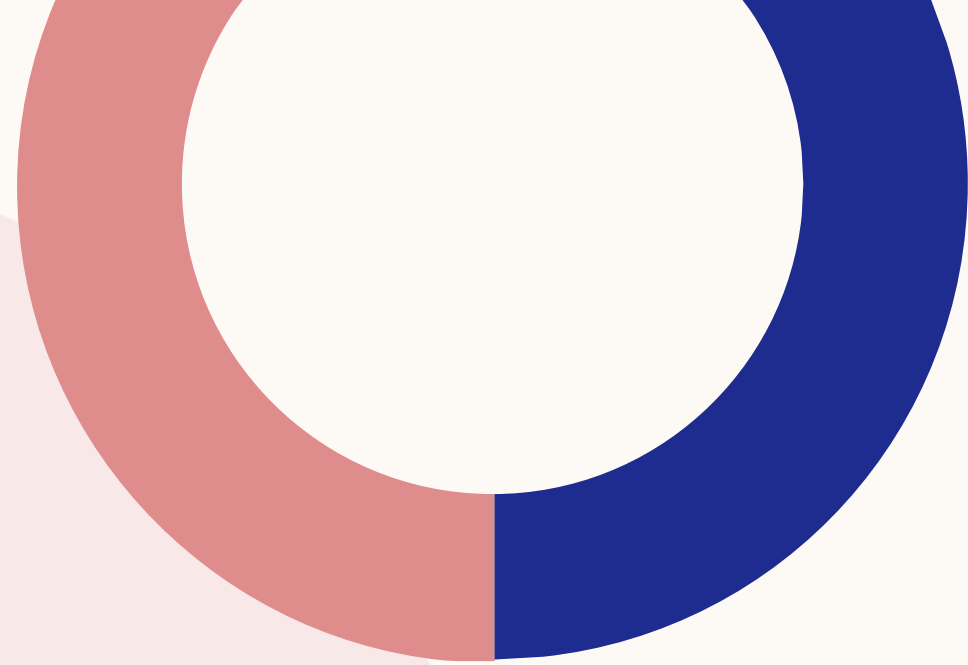


WHAT IS RAID?

- RAID (redundant array of independent disks) is a way of storing the same data in different places on multiple hard disks or solid-state drives (SSDs) to protect data in the case of a drive failure.
- There are different RAID levels, however, and not all aim to provide redundancy.

HOW RAID WORKS?

- RAID works by placing data on multiple disks and allowing input/output (I/O) operations to overlap in a balanced way, improving performance.
- Because using multiple disks increases the mean time between failures, storing data redundantly also increases fault tolerance.



AN IMAGE OF A HARD DRIVE



HOW IT WORKS?

- RAID arrays appear to the operating system (OS) as a single logical drive.
- RAID employs the techniques of disk mirroring or disk striping.
- Mirroring will copy identical data onto more than one drive.
- Striping partitions help spread data over multiple disk drives.
- Each drive's storage space is divided into units ranging from a sector of 512 bytes up to several megabytes.
- The stripes of all the disks are interleaved and addressed in order.
- Disk mirroring and disk striping can also be combined in a RAID array.
- In a single-user system where large records are stored, the stripes are typically set up to be small (512 bytes, for example) so that a **single record spans all the disks and can be accessed quickly by reading all the disks at the same time.**
- In a multiuser system, better performance requires a stripe wide enough to hold the typical or maximum size record, enabling overlapped disk I/O across drives.

RAID CONTROLLER

- A RAID controller is a device used to manage hard disk drives in a storage array. I
- It can be used as a level of abstraction between the OS and the physical disks, presenting groups of disks as logical units.
- Using a RAID controller can improve performance and help protect data in case of a crash.

**““ COMPUTER TECHNOLOGIES ARE
LIKE BUSES. THERE'S ALWAYS
ANOTHER ONE COMING. ””**

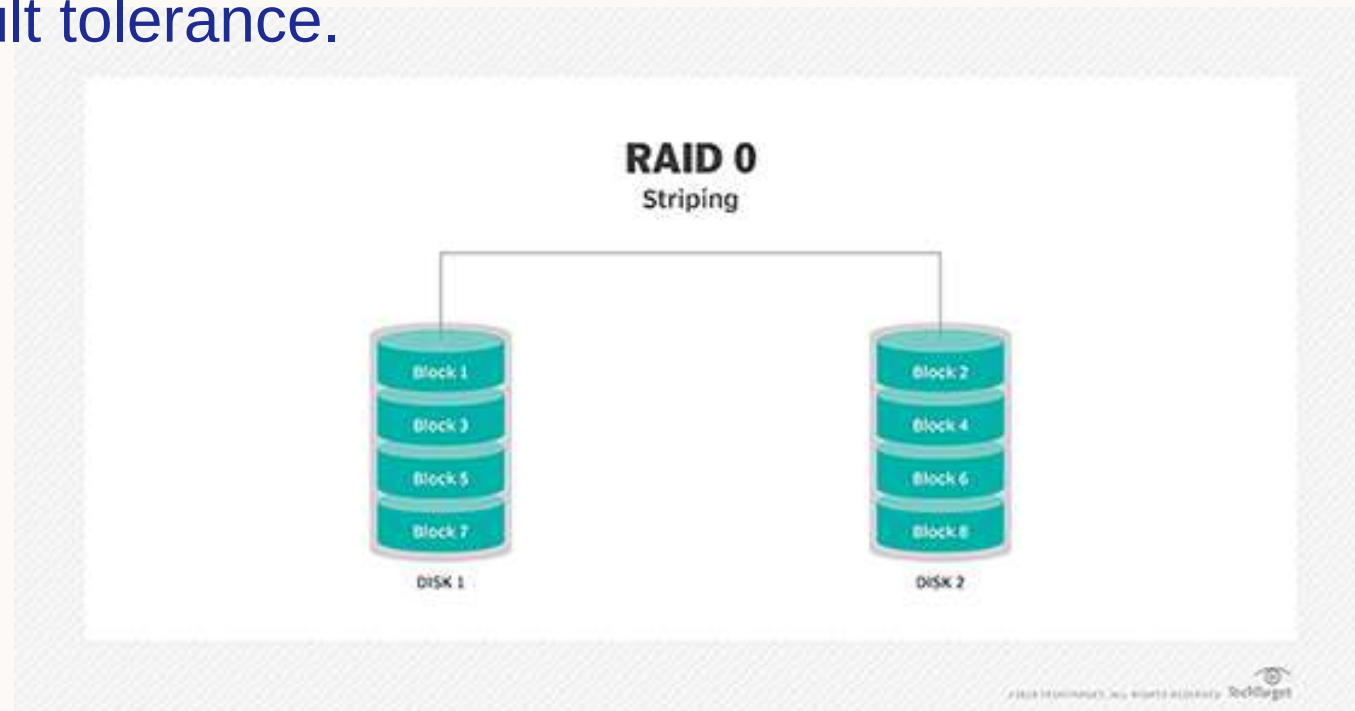
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RAID LEVELS

- RAID devices use different versions, called levels.
- The original paper that coined the term and developed the RAID setup concept defined six levels of RAID -- 0 through 5.
- This numbered system enabled those in IT to differentiate RAID versions.
- The number of levels has since expanded and has been broken into three categories: standard, nested and nonstandard RAID levels.

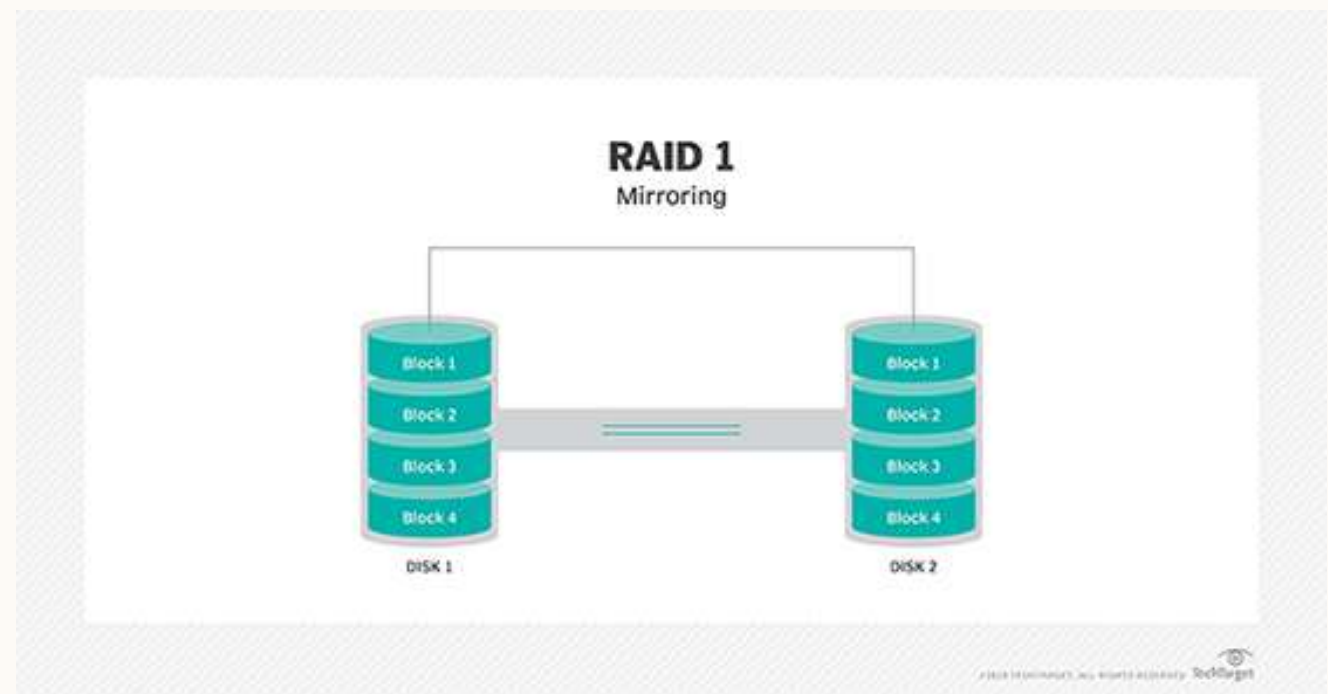
RAID 0

- RAID 0. This configuration has striping but no redundancy of data. It offers the best performance, but it does not provide fault tolerance.



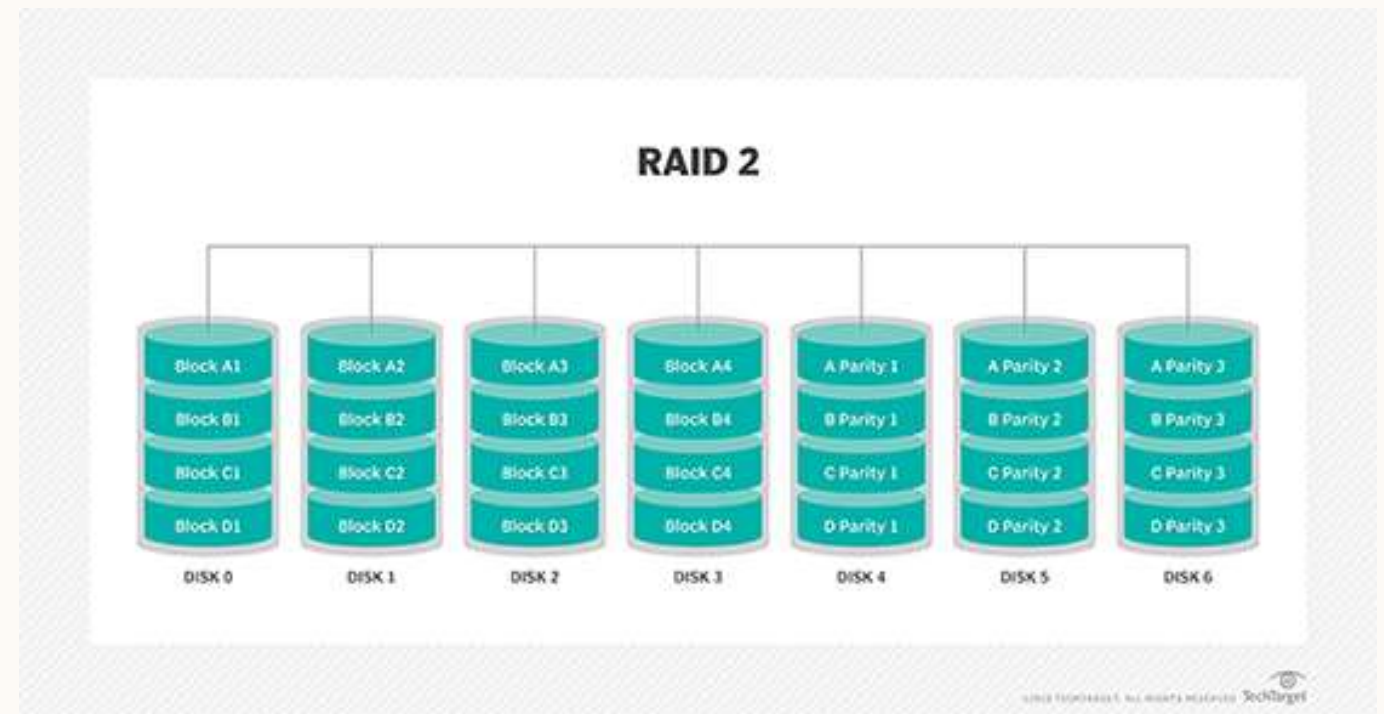
RAID 1

- Also known as **disk mirroring**, this configuration consists of at least two drives that **duplicate the storage of data**. There is no striping. **Read performance is improved** since either disk can be read at the same time. Write performance is the same as for single-disk storage.



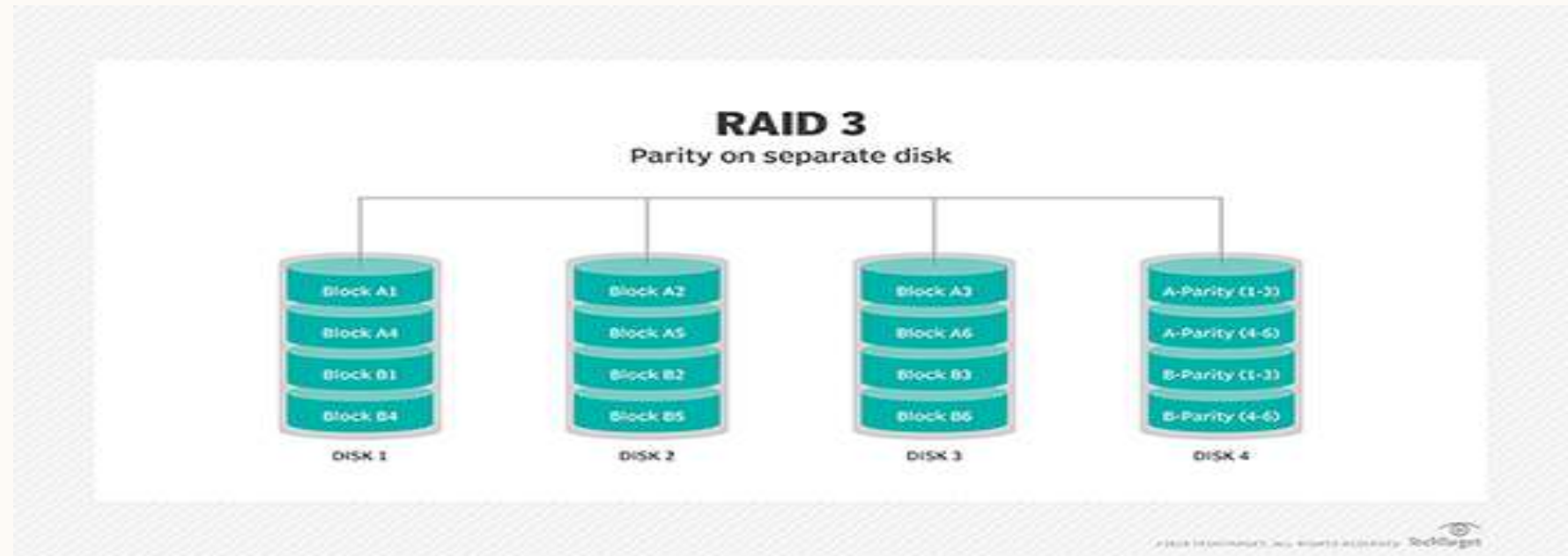
RAID 2

- This configuration uses striping across disks, with some disks storing error checking and correcting (ECC) information. RAID 2 also uses a dedicated Hamming code parity, a linear form of ECC. RAID 2 has no advantage over RAID 3 and is no longer used.



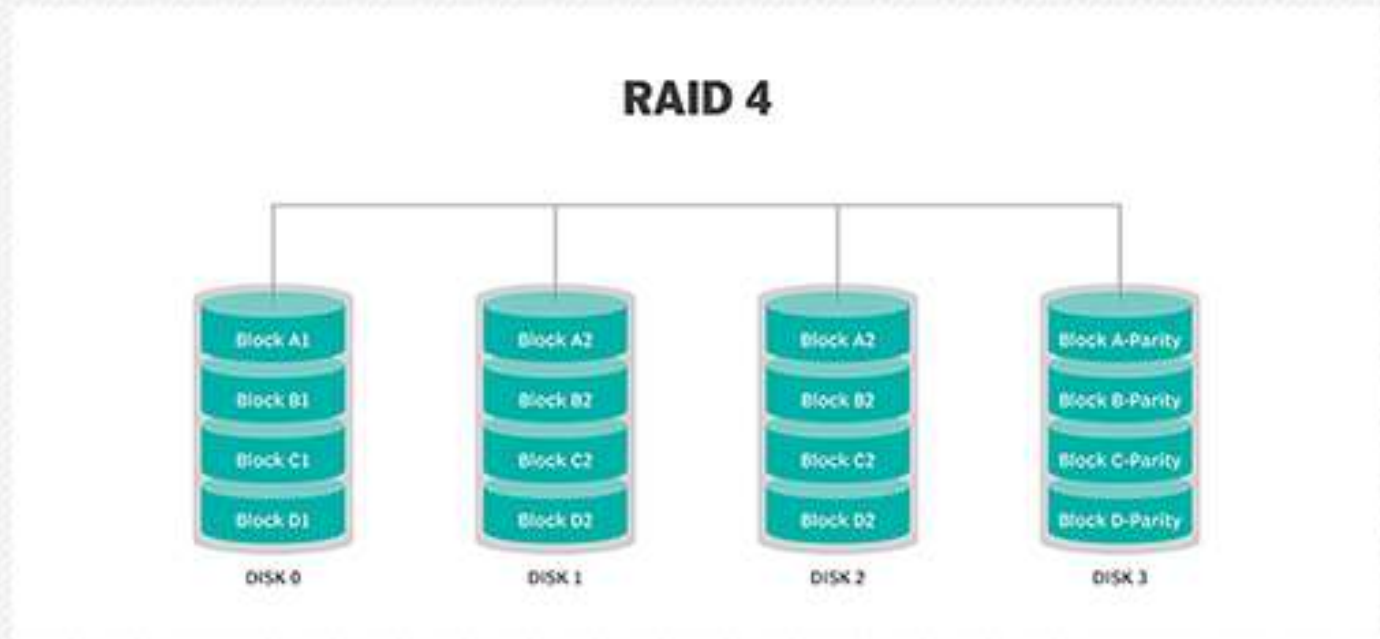
RAID 3

- This technique uses striping and dedicates one drive to **storing parity information**. The embedded **ECC information is used to detect errors**. Data recovery is accomplished by calculating the exclusive information recorded on the other drives. Because an I/O operation addresses all the drives at the same time, RAID 3 cannot overlap I/O. For this reason, RAID 3 is best for single-user systems with long record applications.



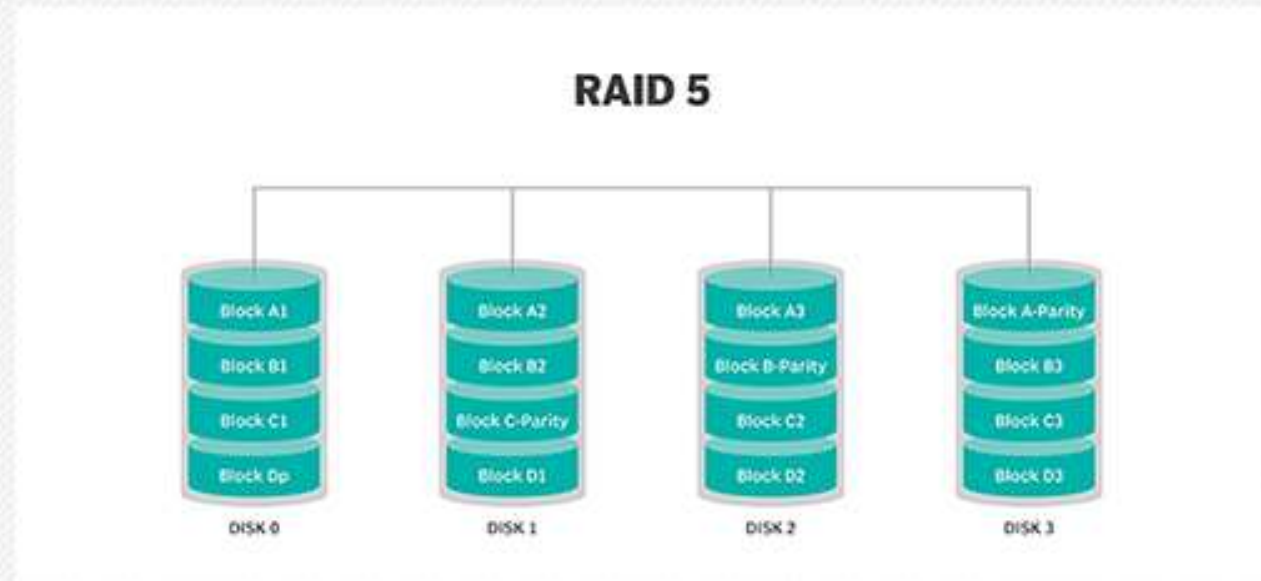
RAID 4

- This level uses large stripes, which means a user can read records from any single drive. Overlapped I/O can then be used for read operations. Because all write operations are required to update the parity drive, no I/O overlapping is possible



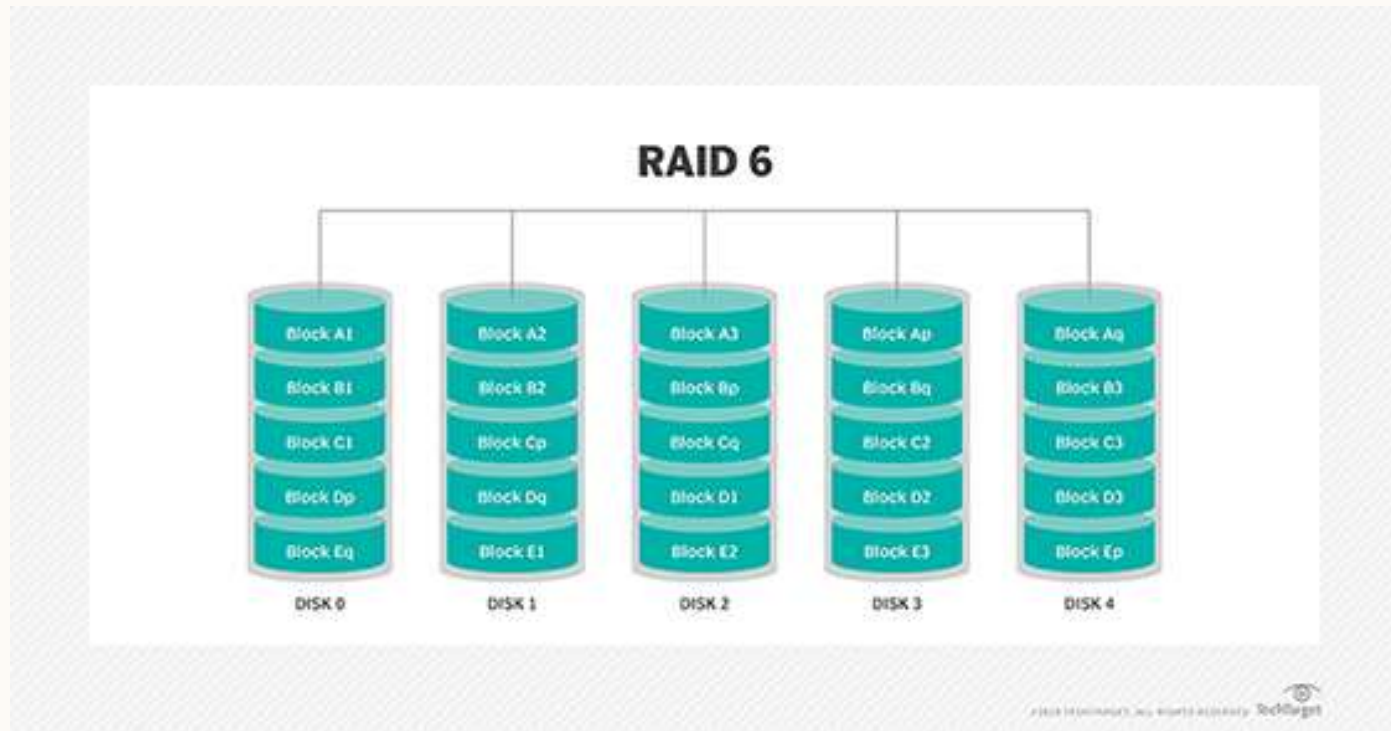
RAID 5

- This level is based on parity block-level striping. The parity information is striped across each drive, enabling the array to function, even if one drive were to fail. The array's architecture enables read and write operations to span multiple drives. This results in performance better than that of a single drive, but not as high as a RAID 0 array. RAID 5 requires at least three disks, but it is often recommended to use at least five disks for performance reasons.



RAID 6

- This technique is similar to RAID 5, but it includes a second parity scheme distributed across the drives in the array. The use of additional parity enables the array to continue functioning, even if two disks fail simultaneously. However, this extra protection comes at a cost. RAID 6 arrays often have slower write performance than RAID 5 arrays.

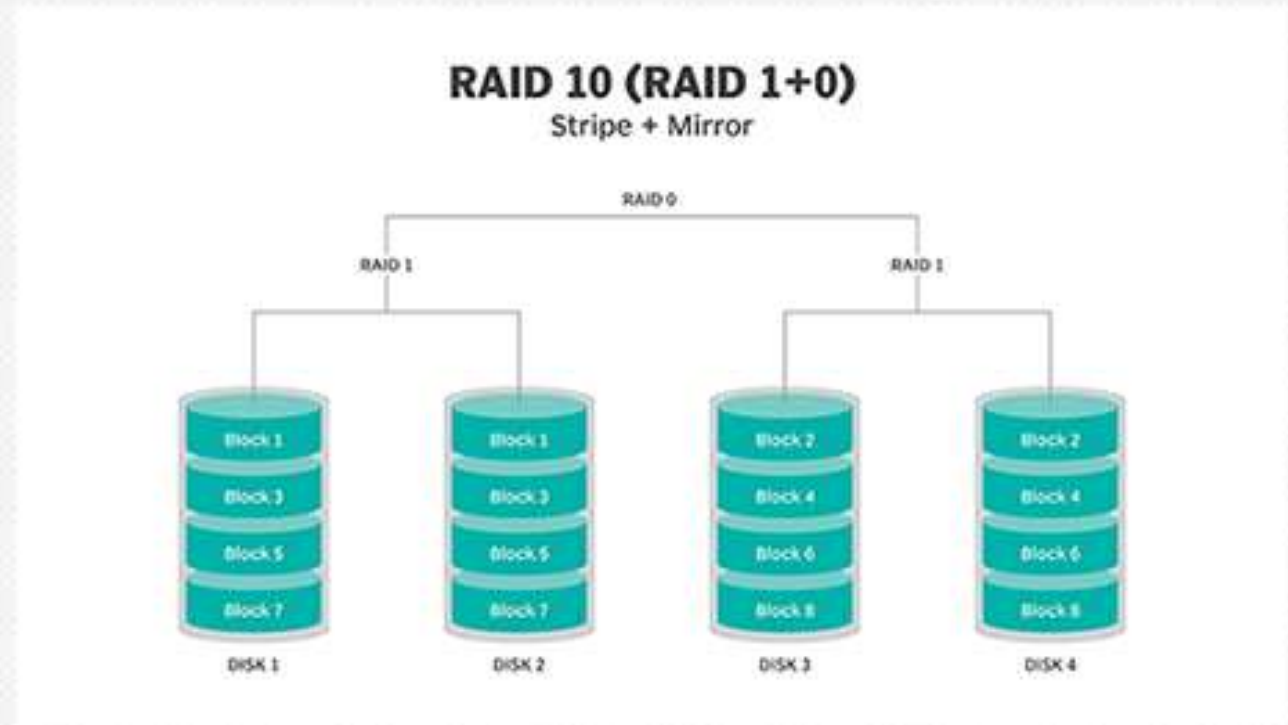


RAID 10 (RAID 1+0). Combining **RAID 1** and **RAID 0**, this level is often referred to as RAID 10, which offers higher performance than RAID 1, but at a much higher cost. In RAID 1+0, the data is mirrored and the mirrors are striped.

NESTED RAID LEVELS

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- Some RAID levels that are based on a combination of RAID levels are referred to as nested RAID. Here are some examples of nested RAID levels.

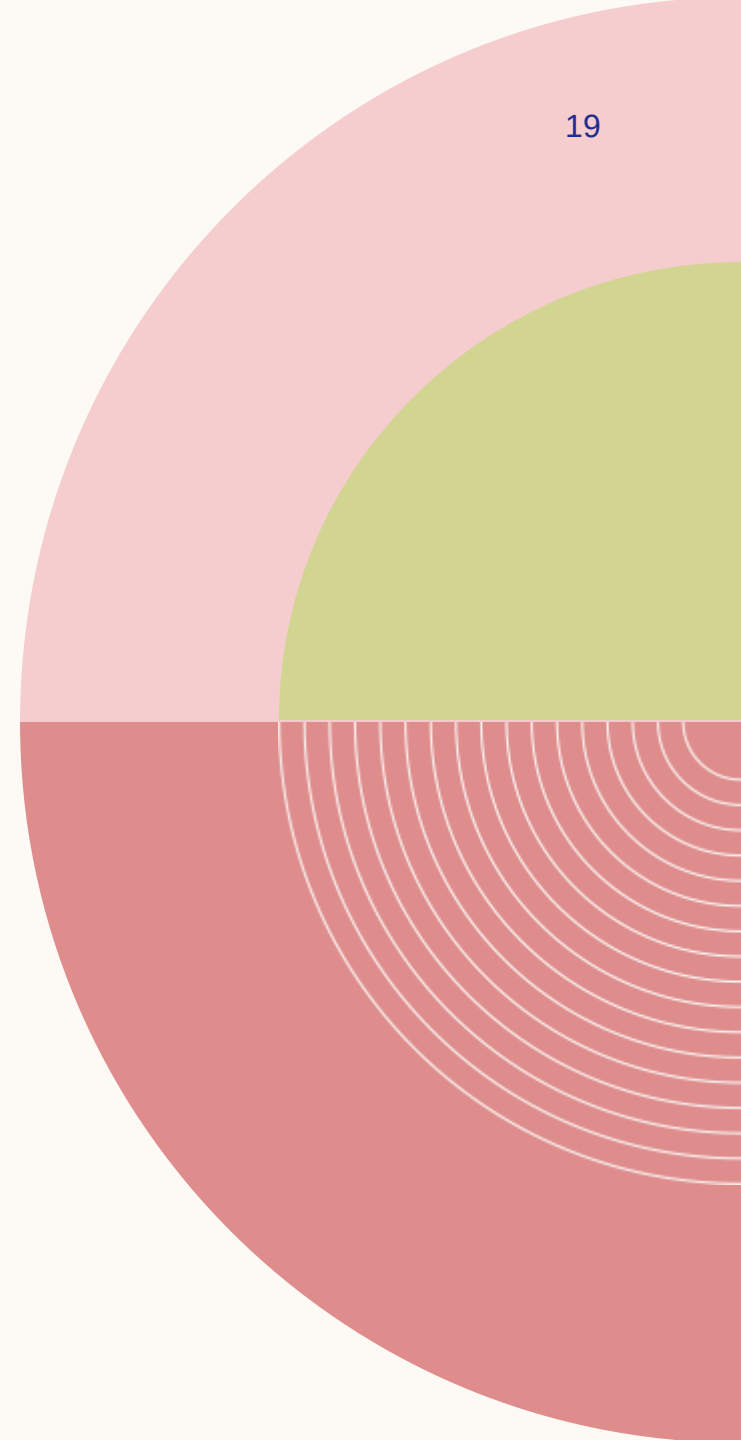
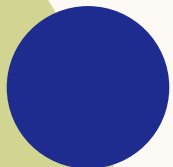


BENEFITS OF RAID

- Improved cost-effectiveness because lower-priced disks are used in large numbers.
- Using multiple hard drives enables RAID to improve the performance of a single hard drive.
- Increased computer speed and reliability after a crash, depending on the configuration.
- Reads and writes can be performed faster than with a single drive with RAID 0. This is because a file system is split up and distributed across drives that work together on the same file.
- There is increased availability and resiliency with RAID 5. With mirroring, two drives can contain the same data, ensuring one will continue to work if the other fails.

SUMMARY

The rise of SSDs is also seen as alleviating the need for RAID. SSDs have no moving parts and do not fail as often as hard disk drives. SSD arrays often use techniques such as wear leveling instead of relying on RAID for data protection. Modern SSDs are fast enough that modern servers may not need the slight performance boost that RAID offers. However, they still may be currently used to prevent data loss.





THANK YOU

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